Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Classical uncertainty principle

Files.

- File: `briefs/ClassicalUncertaintyPrinciple/ClassicalUncertaintyPrinciple.lean`
- Library: `ClassicalUncertaintyPrinciple`
- Imports: four new `ForMathlib` files (the first through the last):
  - `ForMathlib/InformationTheory/GibbsInequality.lean`:
    Gibbs' inequality $-\int \rho \log \rho \le -\int \rho \log G$ for a probability density ρ and a positive G with $\int G \le 1$ (`MeasureTheory.integral_negMulLog_le`), with equality only for $\rho = G$ a.e. (`MeasureTheory.ae_eq_of_integral_negMulLog_eq`)
  - `ForMathlib/Probability/Distributions/Gaussian/Real.lean`: the logarithm of the Gaussian density (`log_gaussianPDFReal`), and its mean and variance as Lebesgue integrals (`integral_mul_gaussianPDFReal`, `integral_sub_sq_mul_gaussianPDFReal`)
  - `ForMathlib/InformationTheory/EntropyVariation.lean`:
    the first variation $\frac{d}{dt}\int -\rho_t \log \rho_t \,\big|_{t=0} = -\int \rho\varphi(\log\rho + 1)$ along $\rho_t = \rho(1+t\varphi)$ for bounded φ (`MeasureTheory.hasDerivAt_integral_negMulLog_mul_one_add`)

– `ForMathlib/InformationTheory/MarginalEntropy.lean`:
the marginal $\rho_X(x) = \int \rho(x,y)\,dy$ of a density of finite entropy on $X \times Y$ has finite entropy if $\rho \log \gamma(y)$ and $\rho \log \gamma'(x)$ are integrable for positive γ, γ' of integral at most 1 (`MeasureTheory.integrable_negMulLog_integral_prod`), and its moments are moments of ρ (`MeasureTheory.integral_mul_integral_prod`).

Setting. An ensemble on a measure space (X, μ) (`IsEnsemble` ρ μ) is a function $\rho \geq 0$ with $\int \rho\,d\mu = 1$ and $\rho \log \rho$ integrable (finite entropy). Its entropy relative to a cell of size c is `entropy` c ρ $\mu = -\int \rho \log(c\rho)\,d\mu$, in nats. A variable $f : X \to \mathbb{R}$ has mean `mean`, variance `var` and standard deviation `sd` under ρ. One DOF has phase space $\mathbb{R} \times \mathbb{R}$ with Lebesgue measure and cell $h > 0$, and $\hbar = h/2\pi$ (`hbar`). The conditions are:

- `LowerEntropicBound` h ρ (ZENT): $S_h[\rho] \geq 0$;
- `UncertaintyPrinciple` h ρ (CUP): $\sigma_q \sigma_p \geq \hbar/e$.

Finite second moments of the variables are hypotheses of the theorems. The Poisson bracket `poissonBracket` f g is $\partial_q f\,\partial_p g - \partial_p f\,\partial_q g$, with Fréchet derivatives. For the brief's own proof, `lagrangian` h λ_1 λ_2 S ρ is $\sigma_q^2\sigma_p^2 + \lambda_1(\int \rho - 1) + \lambda_2(S_h[\rho] - S)$, varied along $\rho(1 + t\varphi)$ with φ bounded and measurable, so $\delta\rho = \rho\varphi$, and `eulerLagrange` is the bracket of its first variation.

For several DOFs the phase space is $(\mathbb{R}^2)^{n+1}$ (`Fin` $(n+1) \to \mathbb{R} \times \mathbb{R}$), and DOF i is the pair (q_i, p_i) of coordinates i. Its marginal `marginal` ρ i is ρ integrated over the other n DOFs. ZENT and CUP become `LowerEntropicBoundPhaseSpace` h ρ ($S_h[\rho_i] \geq 0$ for every i) and `UncertaintyPrinciplePhaseSpace` h ρ ($\sigma_{q_i}\sigma_{p_i} \geq \hbar/e$ for every i, with σ under ρ).

Results.

Brief	Lean
Corollary 1, $S[\rho_U] = \log(\mu(U)/h^n)$	`entropy_uniform` (any measure space, any cell $c > 0$); `entropy_uniform_phaseSpace` (phase space $(\mathbb{R}^2)^n$, $c = h^n$)
Corollary 1, $S[\rho_U] = 0$ iff $\mu(U) = h^n$	`entropy_uniform_eq_zero_iff`
ZENT, CUP	`LowerEntropicBound`, `UncertaintyPrinciple` (one DOF); `LowerEntropicBoundPhaseSpace`, `UncertaintyPrinciplePhaseSpace` (each DOF of $(\mathbb{R}^2)^{n+1}$)
Theorem 2, proof: a product of two independent Gaussians minimizes $\sigma_q\sigma_p$ at given entropy, and then $\sigma_q\sigma_p = (\hbar/e)e^S$	`hbar_div_exp_mul_exp_entropy_le`: $\sigma_q\sigma_p \geq (\hbar/e)e^{S_h[\rho]}$ for every ensemble; `sd_mul_sd_eq_iff`: equality iff ρ is a.e. a product of Gaussians
Theorem 2, eq. (3), ZENT implies CUP	`uncertaintyPrinciple_of_lowerEntropicBound`; for several DOFs `uncertaintyPrinciplePhaseSpace_of_lowerEntropicBoundPhaseSpace`
CUP, equality case	`sd_mul_sd_eq_hbar_div_exp_iff`, corrected (see "Problems found"); for several DOFs `sd_mul_sd_eq_hbar_div_exp_iff_marginal`
ZENT, "$\rho(q,p)$ is the marginal over the DOF"	`isEnsemble_marginal`: the marginal is an ensemble; `mean_marginal`, `var_marginal`: with the means and variances of q_i, p_i under ρ
Eq. (4), the Poisson bracket is the Jacobian determinant	`det_prod`; `poissonBracket_fst_snd`: $\{q,p\} = 1$
Theorem 2, eq. (5), $\sigma_f\sigma_g \geq (\hbar/e)e^{E[\log\lvert\{f,g\}\rvert]}$ under ZENT	`hbar_div_exp_mul_exp_le_sd_mul_sd`; for every ensemble $\sigma_f\sigma_g \geq (\hbar/e)e^{S_h[\rho]+E[\log\lvert\{f,g\}\rvert]}$: `hbar_div_exp_mul_exp_entropy_add_le`
Remark, CUP does not imply ZENT	`exists_not_lowerEntropicBound`
Theorem 2, proof: $\delta L = \int \delta\rho\,[(q-\mu_q)^2\sigma_p^2 + \sigma_q^2(p-\mu_p)^2 + \lambda_1 - \lambda_2\ln(h\rho) - \lambda_2]$	`hasDerivAt_lagrangian`
Theorem 2, proof: the solution of $\delta L = 0$ is a product of two independent Gaussians	`hasDerivAt_lagrangian_zero_iff`: for $\rho > 0$ a.e., $\delta L = 0$ in every bounded direction iff ρ is a.e. the product of Gaussians with the means and variances of ρ, $\lambda_2 = -2\sigma_q^2\sigma_p^2$ and $\lambda_1 = \lambda_2(1 + \ln(\hbar/\sigma_q\sigma_p))$
Theorem 2, proof: then $\sigma_q\sigma_p = (\hbar/e)e^{S[\rho]}$ (used for both equations) the Gaussian bound on any measure space	`sd_mul_sd_eq_of_hasDerivAt_lagrangian` `Hyp.hbar_div_exp_mul_exp_le`: $\sigma_f\sigma_g \geq (\hbar/e)e^{S_h[\rho]+E[\log J]}$ for a weight $J > 0$ that turns every product of Gaussians in (f,g) into a subprobability density (`IsGaussWeight`)

Differences from the brief.

- The bound $\sigma_q\sigma_p \geq (\hbar/e)e^S$ is proved by Gibbs' inequality against the product G of Gaussians in q and p with the means and variances of ρ: $-\int \rho\log\rho \leq -\int \rho\log G = \log(2\pi e\,\sigma_q\sigma_p)$. This proves that the Gaussian is a minimum, not only a critical point, and gives the equality case. Only the marginal variances enter, so no covariance matrix is needed. The brief's Lagrange computation is formalized as well, as a second proof that the Gaussians are the critical points.

- In the Lagrange computation, $\delta\rho$ is $\rho\varphi$ with φ bounded and measurable, rather than arbitrary. Along these variations $\rho(1+t\varphi)$ stays nonnegative for small t and the entropy is differentiable. The vanishing of δL in all of them gives the brief's equation for $\ln(h\rho)$ where $\rho > 0$. The brief solves it for ρ in one line; here the cases $\lambda_2 = 0$ (the quadratic would be constant) and $\lambda_2 > 0$ (ρ would be bounded below, so not integrable) are excluded, and normalization and the variances of ρ fix the constant and λ_2.

- For several DOFs, ZENT and CUP are stated on $(\mathbb{R}^2)^{n+1}$, so there is at least one DOF. An "independent DOF" is read as a pair (q_i, p_i) of coordinates; DOFs in other canonical coordinates, after a change of variables that mixes the pairs, are not considered. The marginal of each DOF is shown to be an ensemble with finite second moments. Its finite entropy comes from Gibbs' inequality in the integrated variables, against a standard Gaussian in the other DOFs, and from the pointwise bound against a standard Gaussian in the DOF itself. Theorem 2 for one DOF then applies to each marginal. Corollary 1 is formalized for any n.

- The hypotheses that the brief leaves implicit are explicit: $\rho \geq 0$, $\int \rho = 1$, finite entropy and finite second moments. Positive variances are proved, not assumed: an ensemble of finite entropy cannot have zero variance in any variable, since Gibbs' inequality against ever narrower Gaussians would bound its entropy by $-\infty$.

- For eq. (5) the brief passes to the density ρ_{fg} of (f, g), with $\rho_{qp} = \rho_{fg}\,|\{f, g\}|$. Here ρ_{fg} is not formed. The Gaussian in (f, g) is pulled back to phase space with the weight $|\{f, g\}|$, and the change-of-variables formula shows that its integral is at most 1. This needs f, g differentiable, $(q, p) \mapsto (f, g)$ injective (not surjective, not orientation preserving), $\{f, g\} \neq 0$ everywhere, and $\rho \log |\{f, g\}|$ integrable. σ_f is the standard deviation of f under ρ. Physical units are not modelled, so the brief's assumption on them is not needed.

- The counterexample for the remark is a uniform distribution on two squares $[L, L+\varepsilon]^2$ and $[-L-\varepsilon, -L]^2$, with $\varepsilon = \sqrt{h}/2$ and $L = \sqrt{h}$. Its entropy is $\log \frac{1}{2} < 0$ and $\sigma_q \sigma_p \geq h/2 \geq \hbar/e$. The brief instead correlates q and p.

Problems found in the brief. The main claim (ZENT implies CUP, eqs. (3) and (5)) is correct; the equality case is misstated.

- CUP says that equality holds "if and only if position and momentum are independent gaussian distributions", and the proof ends the same way. Independent Gaussians satisfy $\sigma_q \sigma_p = (\hbar/e)e^S$, which equals $\hbar/e$ only when $S = 0$. Under ZENT, $\sigma_q \sigma_p = \hbar/e$ holds exactly for products of Gaussians with $\sigma_q \sigma_p = \hbar/e$ (`sd_mul_sd_eq_hbar_div_exp_iff`).

- The Lagrange multipliers give only a stationary point of $\sigma_q^2 \sigma_p^2$ at fixed entropy. They show neither that it is a minimum nor that a minimizer exists, which the proof then uses ("the gaussian is the distribution that minimizes the uncertainty"). The computation also needs $\rho > 0$ almost everywhere, which the brief does not state: a variation cannot move mass onto $\{\rho = 0\}$ while keeping the entropy differentiable, since $-x \ln x$ has infinite slope at 0. So $\delta L = 0$ says nothing about where ρ vanishes (`hasDerivAt_lagrangian_zero_iff` assumes $\rho > 0$ a.e.).

- In the derivation of eq. (5), "order preserving" should be "orientation preserving"; the last two lines of the chain of equalities are identical; and CUP writes $E[\ln\{f, g\}]$ where the abstract and eq. (5) have $E[\ln |\{f, g\}|]$. The logarithm needs the absolute value when $\{f, g\} < 0$.

- The brief does not state that the entropy and the second moments must be finite. Without this $S[\rho]$ or σ may be undefined. For several DOFs, finite entropy of ρ does not even give its marginals finite entropy (not formalized): for any probability density ρ_1 on $\mathbb{R}$, the indicator of $\{(x, y) : |y| < \rho_1(x)/2\}$ has entropy 0 (for $h = 1$) and marginal ρ_1, whose entropy can be $-\infty$. Here the second moment of y is infinite; finite second moments of all DOFs exclude this (`isEnsemble_marginal`).